An independent exact recheck of a claimed finite counterexample to Gale's conjecture in $\mathbb{R}^3$

(a recheck of a result by others; no novelty is claimed)

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Abstract

Frame. V. Voronov (arXiv:2610.28828, 20 Oct 2026) claims a counterexample to Gale's conjecture $\beta_{n,n+1} = \beta_{n+1}(B^n)$ ($n \geq 2$) in $\mathbb{R}^3$: a set V of 101 points (Appendix A, seven decimals) with $\beta_4(V) > 0.88928 > \beta_4(S^2)$. The counterexample is Voronov's; **we claim nothing new and do not claim to have disproved Gale's conjecture.** We re-derived the finite part of his Theorem 2 from the printed coordinates with different code, and checked the ball bound only as far as stated below.

What was rechecked. Exactly (Tier [C]): 303 appendix numbers give 101 distinct points; $D^2 = 100000021020030$ (units 10^{-14}), attained only by one pair; exactly 2030 pairs have distance $> 0.88928\,D$, none on the threshold (weakest included ratio 0.8892809651, strongest excluded 0.8892597950); the 2030-edge distance graph is *not* 4-colourable, with DRAT and LRAT certificates checked by three checkers and a producer/auditor split of encoder, solver and checker; it is 5-colourable; it contains a minimal non-4-colourable subgraph on 425 edges; and $\beta_4(V) = 0.88928096513\ldots$ *exactly* for this rounded set. The configuration is fragile: moving one coordinate by 10^{-3} makes it 4-colourable. Tier [L]: the upper bound $\beta_4(B^3) \leq \sigma_3 = \sqrt{(3+\sqrt{3})/6} = 0.88807$ and the wording of Gale's conjecture are taken from the paper; $\operatorname{diam} T = \sigma_3$ behind the bound is only sampled in floating point (Tier [X]).

What this is NOT. Not a new result; not a proof of anything beyond the finite combinatorial and metric facts above; not a formal proof (no Lean, Tier A not reached); not a check of Gale's 1953 wording, of the optimality of σ_3 for the ball, or of dimensions 4–6 or of the author's search code; not human-audited. The refutation, if it stands, is Voronov's.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code with the argument written down, certificates machine-checked by trusted third-party checkers, not formalised. [X]: floating-point or recorded-absence evidence. Identifiers `GALE-·-nnnn` refer to `ledger/ledger.json`.

1 The claim and its status

For a bounded $A \subset \mathbb{R}^n$ let $\beta_k(A) = \inf \max_i \operatorname{diam} A_i / \operatorname{diam} A$ over partitions into k parts, and $\beta_{n,k} = \max_A \beta_k(A)$ (Voronov, Definitions 1, 2). Gale's conjecture, as quoted by Voronov from Gale (1953): $\beta_{n,n+1} = \beta_{n+1}(B^n)$ for every $n \geq 2$ ([L]; Gale 1953 was not read). Voronov's Theorem 2 [1]: for the 101-point set V of his Appendix A, every partition into four parts has a part of diameter $> 0.88928 \operatorname{diam} V$, while

$\beta_4(S^2) = \sigma_3 = \sqrt{(3+\sqrt{3})/6} \approx 0.88807$ (tetrahedral partition; optimality is credited to Kalai, [L]). The paper states that it has not been independently reviewed and has no Lean formalisation. We found no independent verification, refutation or comment in the literature on 2026-11-01 (alphaXiv lists only the paper and the unrelated-in-content arXiv:2610.39078 [2]; MathWorld restates the claim, which is a report and not a check).

2 Exact geometry ([C])

Extraction. The 303 numbers of Appendix A were extracted by two tools different from the producer's `pdftotext`: PyMuPDF word boxes (clustered into rows, read row-major) and pdfminer.six; both give 303 tokens, equal as multisets, and row-major order equals the author's repository points rounded half-even to seven decimals (maximum rounding error $4.96 \cdot 10^{-8}$). The coordinates are third-party and are *not* reproduced here; `data/third_party_hashes.json` gives hashes and fetch instructions.

Distances (exact rationals). 101 distinct points. $D^2 = 100000021020030 \cdot 10^{-14}$, $D = 1.0000001051$, attained only by the pair $(19, 56)$. With threshold 88928/100000 compared exactly as $\text{dist}^2 > (88928/100000)^2 D^2$ there are exactly 2030 edges and no pair on the threshold. Degrees range over 28..52, the 4-core is all 101 vertices, the maximum clique has size 4 (so non-4-colourability is not from a K_5).

quantity	value	
weakest included pair (rank 2030)	ratio 0.8892809651305	ratio^2 margin $1.7165 \cdot 10^{-6}$, **ratio margin** $9.65 \cdot 10^{-7}$
strongest excluded pair	ratio 0.8892597950	gap to 0.88928: $2.02 \cdot 10^{-5}$
$\sigma_3 = \sqrt{(3+\sqrt{3})/6}$	0.8880738340	$0.88928 > \sigma_3$ exactly: $(6t^2-3)^2 > 3$, $6t^2 - 3 > 0$

Table 1: Extreme ratios ([C]). The first version of the producer's record gave the ratio margin as $9.65 \cdot 10^{-8}$; that was ten times too small (audit correction 1).

Rounding robustness. If every coordinate moves by at most $\delta = 5 \cdot 10^{-8}$ (half a unit in the seventh decimal), each distance and D move by at most $2\sqrt{3}\,\delta \approx 1.7 \cdot 10^{-7}$, so each ratio moves by at most about $(1 + 0.889) \cdot 1.7 \cdot 10^{-7} \approx 3.3 \cdot 10^{-7}$. This is below the included margin $9.65 \cdot 10^{-7}$ and the excluded gap $2.02 \cdot 10^{-5}$: *any* coordinate set that rounds to Appendix A gives the same 2030-edge graph. Direct confirmation: the author's 17-figure repository points give the identical edge set (symmetric difference 0), with a different diameter pair $(12, 37)$ and weakest ratio 0.8892811623939 (the author's stored 0.889281162393).

3 Non-4-colourability ([C])

Producer. Direct encoding (at-least-one, pairwise at-most-one, 2030×4 conflict clauses; 404 variables, 8828 clauses with a symmetry unit for vertex 0), Glucose4 with a DRUP proof, checked by drat-trim (`s VERIFIED`); the same without the symmetry unit (backward and forward drat-trim both `s VERIFIED`); CaDiCaL 1.9.5 on a ladder encoding: UNSAT; own exhaustive DSATUR backtracking: not 4-colourable (745 nodes). An independent parser rebuilt the clause set from the edge list and found both CNF files equal to it.

Auditor (own code, no producer code). A 2-bit encoding (one pair of variables per vertex, four 4-literal clauses per edge, 202 variables, 8120 clauses, no symmetry breaking); encoder controls K_4 (4-colourable), K_5 (not), Moser spindle (4-colourable) and a planted 4-colourable graph with 101 vertices and 2030 edges pass. PicoSAT 0.6.6: UNSAT (also on a direct encoding without at-most-one). CaDiCaL 3.0.1: UNSAT, DRAT proof `s VERIFIED` by drat-trim; LRAT proof verified by `lrat-check.c` (`c VERIFIED`) and by the auditor's own Python LRAT checker (about 40 lines, RUP hints only, RAT rejected; 6246 lemmas). Negative control: with one hint removed from one line both LRAT checkers reject the proof. The producer's no-symmetry proof converted to LRAT by `drat-trim -L` passes both checkers (294 lemmas in the Python checker).

Colourings. The graph is 5-colourable (colouring validated edge by edge), so its chromatic number is 5.

Controls (producer). K_4 ($k = 4$ SAT, $k = 3$ UNSAT), K_5 ($k = 4$ UNSAT), K_6 ($k = 5$ UNSAT), Grötzsch ($k = 3$ UNSAT, $k = 4$ SAT), Moser spindle ($k = 3$ UNSAT, $k = 4$ SAT), Petersen ($k = 3$ SAT, $k = 2$ UNSAT), with all three routes agreeing and every UNSAT proof checked; a bogus or empty proof is rejected by drat-trim; three planted 4-colourable graphs with 101 vertices and 2030 edges are found 4-colourable. (The preregistered expectation "Moser spindle $k = 3$ SAT" was wrong; the spindle has chromatic number 4. Recorded as deviation 1 in `data/RESULT.md`.)

4 The value of $\beta_4(V)$, sweep, criticality, fragility ([C])

Exact value. Sorting all 5050 pairs by length, the 2029 longest pairs form a 4-colourable graph; PicoSAT found a colouring, decoded and validated edge by edge (class sizes 21/20/37/23), whose longest monochromatic pair is exactly pair number 2030. Hence for this rounded set $\beta_4(V) = \sqrt{39541040058709/50000010510015} = 0.88928096513\ldots$ is an *equality*, not merely a lower bound. (The producer had written "$\geq$"; audit correction 3.)

Threshold sweep. Not 4-colourable at $t = 0.888$, 0.888073 (2054 edges), 0.889 (2039), 0.88928 (2030); 4-colourable at 0.889281 (2025 edges) and at $0.8893, 0.8895, 0.890, 0.892, 0.895, 0.90$.

Criticality. A deletion-based search gives a non-4-colourable subgraph with 425 edges on all 101 vertices. The auditor verified exhaustively that all 425 single-edge deletions are 4-colourable (validated colourings), so it is edge-critical; minimum size was not examined. Deleting any one vertex from the full graph leaves a 4-colourable graph (all 101 cases).

Fragility. Moving one coordinate of a random point by 10^{-3} (20 trials) made all 20 graphs 4-colourable (edge counts 1885–2028); 50 random 60-vertex induced subgraphs were all 4-colourable. The configuration is finely tuned, which is consistent with the margin sizes of Section 2 and with an optimised search result.

5 From the graph to the statement (argued, not formalised)

1. V is finite, so $\beta_4(V)$ is a minimum over the finitely many 4-partitions. A 4-partition is a 4-colouring of the vertex set. ([C], argued.)
2. Let G_t have an edge for each pair with distance $> t\,D$. If G_t is not 4-colourable, every 4-partition has a part containing an edge, hence of diameter $> tD$; so $\beta_4(V) > t$ with the ratio taken relative to $D = 1.0000001051$ (the threshold multiplies D, as applied above). With $t = 0.88928$ this is Voronov's claim. ([C], argued.)
3. $0.88928 > \sigma_3$ exactly (Table 1). So $\beta_4(V) > \sigma_3$.
4. Upper bound for the ball ([L]). $\beta_4(B^3) \leq \sigma_3$ follows from the cone partition of the unit-diameter ball over a regular tetrahedron: each cone lies in $\mathrm{conv}(\{0\} \cup T)$ with T a spherical triangle of radius $1/2$, whose diameter is $\max(\operatorname{diam} T, 1/2) = \operatorname{diam} T$; this reduction is rigorous. The chord from a vertex of T to the midpoint of the opposite side is $\frac{1}{2}\sqrt{2 + 2/\sqrt{3}} = \sigma_3$, so $\operatorname{diam} T \geq \sigma_3$ analytically. The equality $\operatorname{diam} T = \sigma_3$ is asserted by the paper and supported here only by dense floating-point sampling (6000 boundary and 4000 interior points, maximum 0.888073833977, equal to σ_3 to $2 \cdot 10^{-16}$): Tier [X], not proved. The *optimality* $\beta_4(B^3) = \sigma_3$ (Kalai) is Tier [L] and is not needed. The tier of this step stays [L].

Steps (1)–(3) concern V only; the statement "$\beta_{3,4} > \beta_4(B^3)$" needs step (4). We stress that the conclusion about Gale's conjecture is Voronov's and rests on step (4) as he states it.

6 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them on the audit's corrections: (1) the ratio margin is $9.65 \cdot 10^{-7}$, not $9.65 \cdot 10^{-8}$; (2) the rounding statement is replaced by the robustness argument of Section 2 (any coordinate set rounding to Appendix A gives the same

graph; the 17-figure set gives the identical edge set); (3) $\beta_4(V)$ is determined exactly for the rounded set; (4) the weakest-edge margin is larger than any rounding effect; (5) the ball bound is reduced rigorously to $\operatorname{diam} T$, which is sampled; `upper_bound_ball` stays [L]; (6) the sampled tetrahedral-triangle value is 0.888073833977 (whole triangle), replacing the producer's coarser 0.8865 sanity value. The remark that the paper's Section 4 says "verivication" is a typo, noted for completeness; whether to contact the author is the owner's decision.

7 Limitations and what is not done

- AI audit only; no human audit. No Lean: the graph-to-β_4 step and the UNSAT statement are not formalised (Tier A not reached).
- Unverified: Gale's 1953 wording (taken from Voronov's quotation); $\operatorname{diam} T = \sigma_3$ (sampled); optimality of σ_3 for the ball (not needed).
- Not done: a `kissat` run (the build was refused by the session's permission classifier); a formally verified checker such as `cake_lpr`; dimensions 4–6; the author's search code.
- Trusted components: drat-trim, `lrat-check.c` (same author family as drat-trim), the auditor's Python LRAT checker, PicoSAT, CaDiCaL 3.0.1, Glucose4 and CaDiCaL 1.9.5 (producer; the proofs are checked, the solvers are not trusted), PyMuPDF, pdfminer.six, Python integers and `fractions`.
- The statements are for the *rounded* Appendix A set (and, by Section 2, every set rounding to it); we did not recertify the author's 17-figure set beyond its edge set.
- Deviations from the preregistration (`PREREG_ITEM.md` sha256 `f3ad4621...c82e278`) are listed in `data/RESULT.md`: the Moser expectation; a control arithmetic slip; two geometry controls run after the headline; a crashed sphere-sampling step replaced by an analytic value plus a sample; a larger threshold list; added checks.

8 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code and derived outputs are in `data/code`, `data/derived`, `data/sat` (CNF and proofs, stored with a `.txt` suffix), `data/logs` (`.log.txt`); the auditor's scripts and small outputs (CNF, DRAT and LRAT proofs, the corrupted-proof control) are in `data/audit`. **The Appendix A coordinates, the raw appendix text, the author's repository CSVs and the PDF are not shipped** (arXiv non-exclusive licence; the author's repository has no licence file); `data/third_party_hashes.json` gives their hashes and fetch instructions, and `data/code/geom.py` regenerates every derived file from the raw appendix text. The edge-index lists and CNF files are derived outputs and contain no coordinates. Inputs are copies of `results/proposal6/2026-11-01/GALE-101/` and of the auditor's scratch directory; nothing was committed, pushed or uploaded.

References

[1] V. Voronov, *A counterexample to Gale's conjecture in* $\mathbb{R}^3$, arXiv:2610.28828 (v1, 20 Oct 2026), unrefereed; Definitions 1, 2, Conjecture 1, Theorem 2, Appendix A. Code: `https://github.com/vsvor/Gale-conjecture`. ([L])

[2] A. Arman, A. Bondarenko, A. Prymak, D. Radchenko, *Borsuk number in four dimensions is at most seven*, arXiv:2610.39078 (29 Oct 2026). Listed by alphaXiv next to [1]; not examined and not used. ([L])

[3] D. Gale, 1953, and J. Kalai's survey, as cited and quoted in [1]; not re-read in the originals. ([L])